SmartRoute Pro — SDA Spatial Clustering Report

Subject: Statistical Data Analysis (SDA) • Equirectangular Metric K-Means & Silhouette Analysis

1. Mathematical Formulation & Equirectangular Metric Projection

Raw spherical coordinates (latitude, longitude) in degrees distort distance calculations at non-equatorial latitudes because meridians converge toward the poles by a factor of cos(latitude). To calculate geometrically meaningful Euclidean distances in kilometers without expensive geodesic transforms on local scales, coordinates are projected onto a local metric tangent plane with origin at (mean_lat, mean_lon):

• x_km = (lon - mean_lon) * 111.320 * cos(mean_lat * pi / 180)

• y_km = (lat - mean_lat) * 110.574

K-Means Lloyd's algorithm partitions N locations into K clusters by iteratively minimizing the Within-Cluster Sum of Squares (Inertia): J = sum_{k=1}^K sum_{x in S_k} ||x - mu_k||^2.

2. Cluster Partitioning & Silhouette Quality Evaluation

Metric	Evaluated Value
Cluster Run ID	**Run #4**
Cluster Count (K)	**3**
Total Locations Clustered	7
WCSS / Inertia	**14.8075**
Silhouette Coefficient	**0.2815**
Iterations to Convergence	3
Computation Runtime	**18.55 ms**
Spatial Projection	Equirectangular Local Metric Approximation (km)

3. Partition Centroids & Radius Distribution

Cluster #	Centroid Lat	Centroid Lon	Assigned Nodes	Cluster Radius	Mean Node Distance
Cluster 0	37.8100	-122.4200	**1**	0.00 km	0.00 km
Cluster 1	37.7822	-122.4046	**4**	1.57 km	1.20 km
Cluster 2	37.7535	-122.4400	**2**	2.05 km	2.05 km